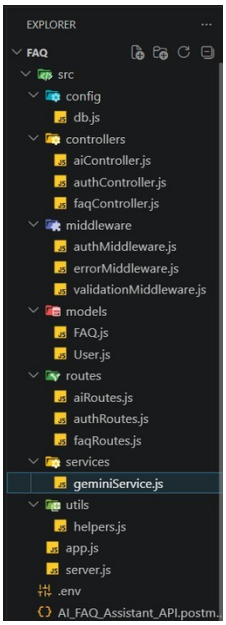
Code-Layout, Readability and Reusability

Date	25 September 2026
Team ID	SWTID-2026-3976
Project Name	AI FAQ Assistant API (AnswerBot)
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

The source code follows a layered, MVC-style structure under **src/**: config, controllers, middleware, models, routes, services and utils, with app.js and server.js as entry points.



Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing used throughout the code	Yes	Two-space indentation across all files
2	Proper File Structure	Files and folders are logically organised	Yes	Separate folders for routes, controllers, services, models, middleware
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	e.g. faqSchema, registerUser, protect
4	Function / Method Names	Functions are descriptively named	Yes	e.g. createFAQ, searchFAQs, generateAIFAQ
5	Code Comments	Inline and block comments explain logic	Partial	Key routes and middleware commented; more comments possible
6	Modular Design	Code is split into reusable functions/modules	Yes	Routes, controllers and Gemini service are independent modules

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
7	No Redundant Code	Duplicate or unused code is removed	Yes	Shared logic kept in middleware and utils/helpers.js
8	Error Handling	Exceptions and errors are handled gracefully	Yes	Central errorMiddleware and unhandledRejection handler

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1	authMiddleware (protect)	JavaScript / Express	Profile, FAQ write routes and AI routes	High
2	validationMiddleware	JavaScript / Express	Auth, FAQ and AI routes	High
3	errorMiddleware	JavaScript / Express	Whole application	High
4	geminiService	JavaScript / @google/genai	AI answer and AI FAQ generation	Medium
5	db.js (Mongoose connection)	JavaScript / Mongoose	server.js start-up	Medium

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	4	Clear layered folders and entry points
Readability	4	Descriptive names and consistent formatting
Reusability	4	Middleware and services are shared across routes
Documentation / Comments	3	README and report available; inline comments can be extended
Overall Score	3.75	Average of the four ratings